

A comparative study of five classical nonlinear models for kinetic growth modelling of *Dunaliella* sp.

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Abstract: Mathematical modelling of microalgal growth is a fundamental tool for describing biomass dynamics and supporting the analysis and optimisation of cultivation processes. In this study, the batch growth of the halotolerant microalga *Dunaliella* sp. was investigated using five classical nonlinear growth models: exponential, logistic, Gompertz, Richards, and Baranyi. The experimental growth profile exhibited a clear sigmoidal pattern characterised by an adaptation phase, rapid biomass accumulation, and a transition to stationary conditions. Model parameters were estimated using nonlinear least-squares regression based on the Levenberg–Marquardt algorithm. Model evaluation combined descriptive and information-theoretic criteria, including root mean square error (RMSE), bias factor (BF), residual diagnostics, Akaike’s information criterion corrected for small sample size (AICc), $\Delta AICc$, and Akaike weights.

The results indicated that flexible sigmoidal formulations provided a more realistic representation of the observed growth dynamics. Specifically, the logistic model received the strongest statistical support based on AICc while maintaining parameter parsimony. The Richards model remained statistically plausible but did not provide a sufficiently better fit to the experimental data to justify its additional complexity. These findings demonstrate the value of rigorous information-theoretic model selection. They also provide a robust framework for predictive modelling of *Dunaliella* sp. growth, supporting improved process design, operational decision-making, and scale-up strategies in microalgal biotechnology.

Keywords: *Dunaliella* sp., growth, kinetic modelling, Levenberg–Marquardt method, statistics

INTRODUCTION

Microalgae are photosynthetic microorganisms with high potential for biotechnological applications, particularly in the fields of biofuels, pharmaceuticals, and wastewater treatment. Microalgae cultivation requires relatively little land, and its low operational costs make this biotechnological alternative a cost-effective, environmentally friendly, and sustainable solution for obtaining bioproducts (Laizu, Hassan and Ahmad, 2025). Microalgae-based

systems utilise sunlight as their main energy input through photoautotrophic metabolism, offering a more energy-efficient alternative to conventional technologies. In addition to capturing atmospheric carbon dioxide, these systems enable the production of high-value biomass, thereby supporting the development of a sustainable microalgal circular economy (Mehariya *et al.*, 2021; Piyatilleke *et al.*, 2024).

Among the wide diversity of microalgal species, *Dunaliella* sp. stands out because of its exceptional tolerance to high salinity

and environmental stressors, as well as its capacity to accumulate high-value metabolites such as carotenoids, lipids, carbohydrates, and proteins (Tan *et al.*, 2025). These characteristics make it a promising candidate for industrial applications, including food, animal feed, cosmetics, agriculture, pharmaceuticals, and energy (Bhat, Unpaprom and Ramaraj, 2024; Tambat *et al.*, 2025). Consequently, understanding and accurately modelling the growth behaviour of *Dunaliella* is essential for the design, optimisation, and scale-up of biotechnological processes involving this genus.

Kinetic models are fundamental tools for elucidating microalgal growth dynamics. They enable the prediction of biomass accumulation under varying conditions and help clarify the factors influencing growth. Consequently, their application supports the optimisation of bioreactor configurations, cultivation protocols, and process management, fostering more sustainable production practices (Esteves *et al.*, 2024). The most commonly used models include the exponential, logistic, Gompertz, Richards, and Baranyi models (Baranyi and Roberts, 1994; Sousa *et al.*, 2021). Each model offers different advantages depending on the shape of the growth curve and the presence of distinct phases, such as lag, exponential, and stationary phases. However, selecting the most appropriate model depends on its fit to experimental data and its ability to capture key biological phenomena, such as adaptation time and growth inhibition (Mohd, Yasin and Takriff, 2021).

Despite the extensive application of these models in microalgal studies, comparative kinetic analyses focusing specifically on *Dunaliella* sp. remain limited. Most previous investigations apply a single growth model or evaluate different models under varying experimental conditions. This complicates direct comparisons and limits the generalisation of conclusions (Sohrabi, Jazini and Shariati, 2019). Furthermore, model selection in many studies relies primarily on conventional goodness-of-fit metrics, such as the coefficient of determination (R^2). Such metrics are known to be inadequate for comparing nonlinear models with different numbers of parameters and may lead to ambiguous or misleading conclusions.

To address these limitations, systematic comparative studies are needed to evaluate multiple growth models under identical experimental conditions using statistically robust criteria. Information-theoretic approaches, such as Akaike's information criterion corrected for small sample size (AICc), provide an objective basis for comparing nonlinear models by balancing goodness-of-fit with model parsimony. These approaches have been increasingly recommended in ecological and bioprocess modelling (Esteves *et al.*, 2024).

Although classical sigmoidal models are widely applied in microalgal research, direct comparative evaluations of exponential, logistic, Gompertz, Richards, and Baranyi formulations under identical experimental conditions remain scarce for *Dunaliella* species. Moreover, few studies integrate information-theoretic criteria, bias analysis, and residual diagnostics within a unified nonlinear regression framework. Therefore, a statistically rigorous comparison that explicitly addresses model complexity and structural adequacy is warranted. By combining these analytical components within a single dataset, the present study advances methodological standards in microalgal growth modelling. It also clarifies the relative suitability of widely used kinetic formulations for halotolerant microalgae.

The aim of the present study is to systematically compare five widely used nonlinear growth models (exponential, logistic,

Gompertz, Richards, and Baranyi) for describing the batch kinetics of *Dunaliella* sp. cultivated under controlled laboratory conditions. Using nonlinear regression combined with information-theoretic model selection (AICc), bias factor analysis, and residual diagnostics, this study seeks to identify the most statistically supported and biologically consistent formulation. By integrating goodness-of-fit metrics with an assessment of model complexity within a unified framework, this work contributes to improved methodological rigour in microalgal growth modelling. It also provides guidance for reliable parameter estimation in bioprocess design.

MATERIALS AND METHODS

EXPERIMENTAL CONDITIONS

The microalga *Dunaliella* sp. was obtained from the biotechnology laboratory of the University of Sinu, located in Cartagena de Indias, Colombia. The strain was identified morphologically at the genus level and is therefore referred to as *Dunaliella* sp. Molecular characterisation (e.g., sequencing and a National Center for Biotechnology Information – NCBI accession number) was not performed in the present study.

Cultivation was carried out in batch mode for 39 days under controlled laboratory conditions using a stepwise scale-up procedure from Petri dishes to test tubes, then to 250 cm³ Erlenmeyer flasks, and finally to 4,000 cm³ flasks. *Dunaliella* sp. strains were cultured in Conway medium containing the following components: FeCl₃·6H₂O (26 g), MnCl₂·4H₂O (0.72 g), H₃BO₃ (67.2 g), EDTA (90 g), Na₂HPO₄·12H₂O (40 g), NaNO₃ (200 g), Na₂SiO₃ (40 g), H₂O (2 dm³), 2 cm³ of a trace-metal solution prepared with ZnCl₂ (2.1 g), CoCl₂·6H₂O (2 g), (NH₄)₆Mo₇O₂₄·4H₂O (0.9 g), CuSO₄·5H₂O (2 g), and distilled water (100 cm³), and 100 cm³ of a vitamin solution composed of decamyl (210 mg) and distilled water (100 cm³) (Baldiris-Navarro *et al.*, 2023).

Cultures were grown under white LED illumination (~100 μmol photons·m⁻²·s⁻¹) with a 16 h-light/8 h-dark photoperiod. Aeration was supplied through filtered air enriched with 2% CO₂. The temperature was maintained at 25 ± 1°C. The pH was not actively controlled but was monitored daily using a calibrated pH meter. During cultivation, pH values remained within the range of 7.5–8.5. Cell density was monitored daily by direct microscopic counting using a Neubauer chamber. Optical density was measured at 680 nm using an ultraviolet-visible (UV-Vis) spectrophotometer. All growth experiments were performed in biological triplicate, and the results are reported as mean ± standard deviation.

KINETIC MODELS AND PARAMETERS

Microalgal growth in batch culture generally exhibits four distinct phases: lag, exponential, stationary, and decline (Ramírez Duque, 2017). Thus, the mathematical model needs to be robust enough to simulate these different growth phases. To describe these dynamics, five widely used nonlinear growth models were evaluated: exponential, logistic, modified Gompertz, Richards, and Baranyi–Roberts (Tab. 1).

Table 1. Microalgal kinetic model equations

Model	Equation	Explanations
Exponential	$X(t) = X_0 \cdot e^{\mu t}$ (1)	X = biomass concentration at time t , X_0 = initial biomass, e = Euler's number, μ = specific growth rate, t = time
Logistic	$X(t) = \frac{X_{\max}}{1 + \left(\frac{X_{\max} - X_0}{X_0}\right) e^{-\mu t}}$ (2)	$X_{\max}$ = maximum biomass
Gompertz (modified)	$X(t) = X_{\max} \cdot \exp\left[-\exp\left(\frac{\mu e}{X_{\max}}(\lambda - t) + 1\right)\right]$ (3)	λ = duration of the lag phase
Richards	$X(t) = X_{\max}(1 + a \cdot e^{-kt})^{-1/\nu}$ (4)	a = integration constant related to the initial condition, k = growth rate constant, ν = dimensionless shape parameter controlling curve asymmetry
Baranyi-Roberts	$X(t) = X_0 + \frac{X_{\max} - X_0}{1 + \exp[-\mu(A(t) - h_0)]}$ (5)	$A(t)$ = physiological adjustment function describing cell adaptation

Source: own elaboration.

The exponential growth model is one of the simplest and most fundamental models used to describe microbial population dynamics. It assumes that the specific growth rate (μ) remains constant over time and that environmental resources such as nutrients, light, and space are not limiting. However, it fails to account for the lag phase, nutrient depletion, waste accumulation, or the onset of the stationary and decline phases.

The logistic model is one of the most widely used models for describing microbial growth in closed systems because it adequately represents the transition between the exponential and stationary phases. This model assumes that population growth is limited by environmental factors, such as nutrients or space. This limitation is reflected in a sigmoidal curve that reaches a maximum biomass or carrying capacity (Kassim, Ramli and Meng, 2022). The Gompertz model is an empirical sigmoidal growth model frequently used in microbiology to describe population dynamics with an initial lag phase. In this model, $X_{\max}$ is the maximum biomass, $\mu_{\max}$ is the maximum specific growth rate, and λ represents the duration of the lag phase. One of its strengths is its ability to capture delayed growth due to cellular adaptation (Perez-Guzman *et al.*, 2024; Wang and Guo, 2024).

The Richards model is a flexible, generalised sigmoidal growth model that includes an additional shape parameter to allow for variable curve asymmetry. In this model, $X_{\max}$ is the maximum biomass, while a , k , and ν are model-specific constants that determine the growth rate and curve shape. Unlike the logistic and Gompertz models, Richards' equation can adapt to a wide variety of growth behaviours, making it especially useful for datasets that do not conform to standard sigmoidal forms (Ma and Jian, 2023). The Baranyi model is a mechanistic growth model that explicitly incorporates the physiological state of the cells during the lag phase. It uses a time-dependent function $A(t)$ to describe the transition from adaptation to exponential growth. Here, X_0 is the initial biomass, $X_{\max}$ is the maximum biomass, μ is the specific growth rate, and h_0 quantifies the duration of the adaptation period (Baranyi and Roberts, 1994).

All model parameters were estimated through nonlinear regression, and the mathematical formulations of the evaluated models are summarised in Table 1. Parameter interpretation was

conducted in conjunction with statistical model-selection criteria to ensure that the conclusions were supported by both quantitative performance and biological plausibility.

MODEL FITTING AND STATISTICAL EVALUATION

Nonlinear regression was performed using the Levenberg-Marquardt algorithm (LM), which minimises the sum of squared residuals between observed data and model predictions. The Levenberg-Marquardt algorithm, also known as the damped least-squares method, is specifically designed for error functions expressed as sums of squared errors.

Assuming that the error function may be formulated as:

$$f(\omega) = \sum_{i=0}^m e_i^2(\omega) \quad (6)$$

where: m = the size of the data set, $e_i(\omega)$ = the error that is made in the i -th instance, ω = the parameter vector.

The LM method iteratively updates the parameter vector according to:

$$\omega_{i+1} = \omega_i - (J_i^T J_i + \lambda_i I)^{-1} \cdot (J_i^T e_i) \quad (7)$$

where: J = the Jacobian matrix of partial derivatives, I = the identity matrix, λ = a damping factor that controls the step size, $J_i^T J_i$ approximates the Hessian matrix of the objective function.

When $\lambda = 0$, Newton's method is obtained using the approximate Hessian. If λ_i is very large, the method becomes a gradient descent algorithm with a small learning rate. Therefore, λ_i is initialised at a large value so that the first update is a small step in the gradient descent direction. If an iteration fails, λ is increased by a certain factor. Otherwise, as the error decreases, λ decreases, and the Levenberg-Marquardt algorithm approaches Newton's method. Convergence was assumed when the relative variation in parameters was below 10^{-6} between successive iterations (Ling, Jamaian and Zulkifli, 1988; Hecke van, 2017; Yan *et al.*, 2021; Gavin, 2026).

For each growth model, parameter estimation was performed by minimising the sum of squared residuals between experimentally observed cell concentrations and model-predicted values. Residuals were defined as the differences between measured and predicted biomass concentrations at each sampling time. Model fitting was conducted using the full experimental dataset obtained from batch cultivation. Convergence was verified by monitoring changes in parameter values and residual error across iterations.

Model performance was first evaluated using the root mean square error (*RMSE*) as a descriptive measure of fitting error. Because *RMSE* was calculated on the same dataset used for parameter estimation, it was interpreted strictly as a goodness-of-fit metric rather than as a measure of predictive accuracy. Systematic deviation was assessed using the bias factor (*BF*), with values close to unity indicating unbiased model performance (Abbas, Ni and Du, 2024; Abbas, Ni and Du, 2025).

To enable rigorous comparison among nonlinear models with different numbers of parameters, information-theoretic model-selection criteria were applied. Akaike's information criterion corrected for small sample size (*AICc*) was calculated for each model according to standard formulations. Relative model support was assessed using the difference in *AICc* values ($\Delta AICc$) between each model and the model with the minimum *AICc*, as well as Akaike weights (w_i). These weights represent the relative probability that a given model is the best among the set of candidate models (Burnham and Anderson, 2002) (Tab. 2).

Model selection was therefore based primarily on *AICc*, $\Delta AICc$, and w_i , which balance goodness-of-fit and model parsimony and are appropriate for nonlinear regression analysis. Conventional metrics such as the coefficient of determination (R^2) were not used as decisive criteria because of their known limitations when comparing nonlinear models with different

parameter complexities. Statistical metrics were interpreted in conjunction with biological plausibility to ensure a meaningful representation of *Dunaliella* growth dynamics.

Nonlinear regression and statistical analyses were conducted using Microsoft Excel and RStudio Version 2025.09.0, with standard nonlinear least-squares routines.

RESULTS

The batch cultivation of *Dunaliella* sp. over a 39-day period is shown in Figure 1. After a short lag phase during the first two days, a pronounced exponential growth phase was observed between days 2 and 20. This was followed by clear growth

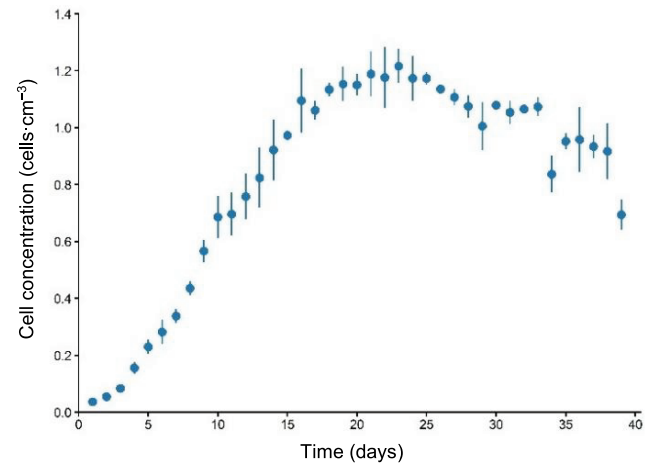


Fig. 1. *Dunaliella* sp. growth profile; data are presented as mean $\pm$ standard deviation ($n = 3$ biological replicates); source: own study

Table 2. Statistical metrics used for model fitting and comparison

Statistic	Equation	Description
Residual (e_i)	$e_i = y_i - \hat{y}_i$ (8)	difference between the experimental observation (y_i) and the model-predicted value ($\hat{y}_i$) at time (i)
Sum of squared errors (<i>SSE</i>)	$SSE = \sum (y_i - \hat{y}_i)^2$ (9)	total squared deviation between experimental and predicted values
Root mean square error (<i>RMSE</i>)	$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$ (10)	descriptive measure of the average magnitude of fitting errors
Bias factor (<i>BF</i>)	$BF = 10^{\frac{1}{n} \sum \log\left(\frac{\hat{y}_i}{y_i}\right)}$ (11)	measures systematic deviation between predicted and observed values
Akaike information criterion (<i>AIC</i>)	$AIC = n \cdot \ln\left(\frac{SSE}{n}\right) + 2k$ (12)	information criterion balancing model fit and complexity, where n = the number of observations, k = the number of model parameters
Corrected Akaike information criterion (<i>AICc</i>)	$AICc = AIC + \frac{2k(k+1)}{n-k-1}$ (13)	<i>AIC</i> corrected for small sample size; recommended when $n/k < 40$
Difference in <i>AICc</i> ($\Delta AICc$)	$\Delta AICc_i = AICc_i - AICc_{\min}$ (14)	difference between the <i>AICc</i> of model i and the minimum <i>AICc</i> among all candidate models
Akaike weight (w_i)	$w_i = \frac{e^{-0.5\Delta AICc_i}}{\sum_{j=1}^m e^{-0.5\Delta AICc_j}}$ (15)	relative probability that model i is the best-supported model among the candidate set of m models

Source: own elaboration.

deceleration and a stationary phase towards the end of the cultivation period. Experimental variability was assessed using biological triplicates. The coefficient of variation (CV) ranged from approximately 3.3% to 14.3% throughout cultivation, with a mean value of about 8–9%. Variability was higher during the initial low-density phase and decreased during the exponential and stationary phases, indicating acceptable experimental reproducibility.

Cell concentrations increased from initial values on the order of 10^5 cells·cm⁻³ to maximum values exceeding $1.1 \cdot 10^7$ cells·cm⁻³ (Fig. 1). The data represent the mean cell concentration $\pm$ standard deviation (SD) of three biological replicates ($n = 3$).

The growth data for *Dunaliella* sp. were used to determine the parameter values for each model. The Levenberg–Marquardt (LM) algorithm was applied to each model. Estimated kinetic parameters for each model are summarised in Table 3.

The comparison between experimental and estimated data for each kinetic model is shown graphically in Figure 2. Model performance was first evaluated using the root mean square error

Table 3. Fitting coefficients for all kinetic models

Model	$\mu_{\max}$ (day ⁻¹)	$X_{\max}$ ($\cdot 10^6$ cells·cm ⁻³)	λ (day)
Exponential	0.38	–	0.0
Logistic	0.25	1.85	1.2
Gompertz	0.22	1.80	1.5
Richards	0.30	1.90	1.4
Baranyi	0.27	1.82	1.3

Explanations: $\mu_{\max}$ = maximum specific growth rate, $X_{\max}$ = maximum biomass, λ = duration of the lag phase.

Source: own study.

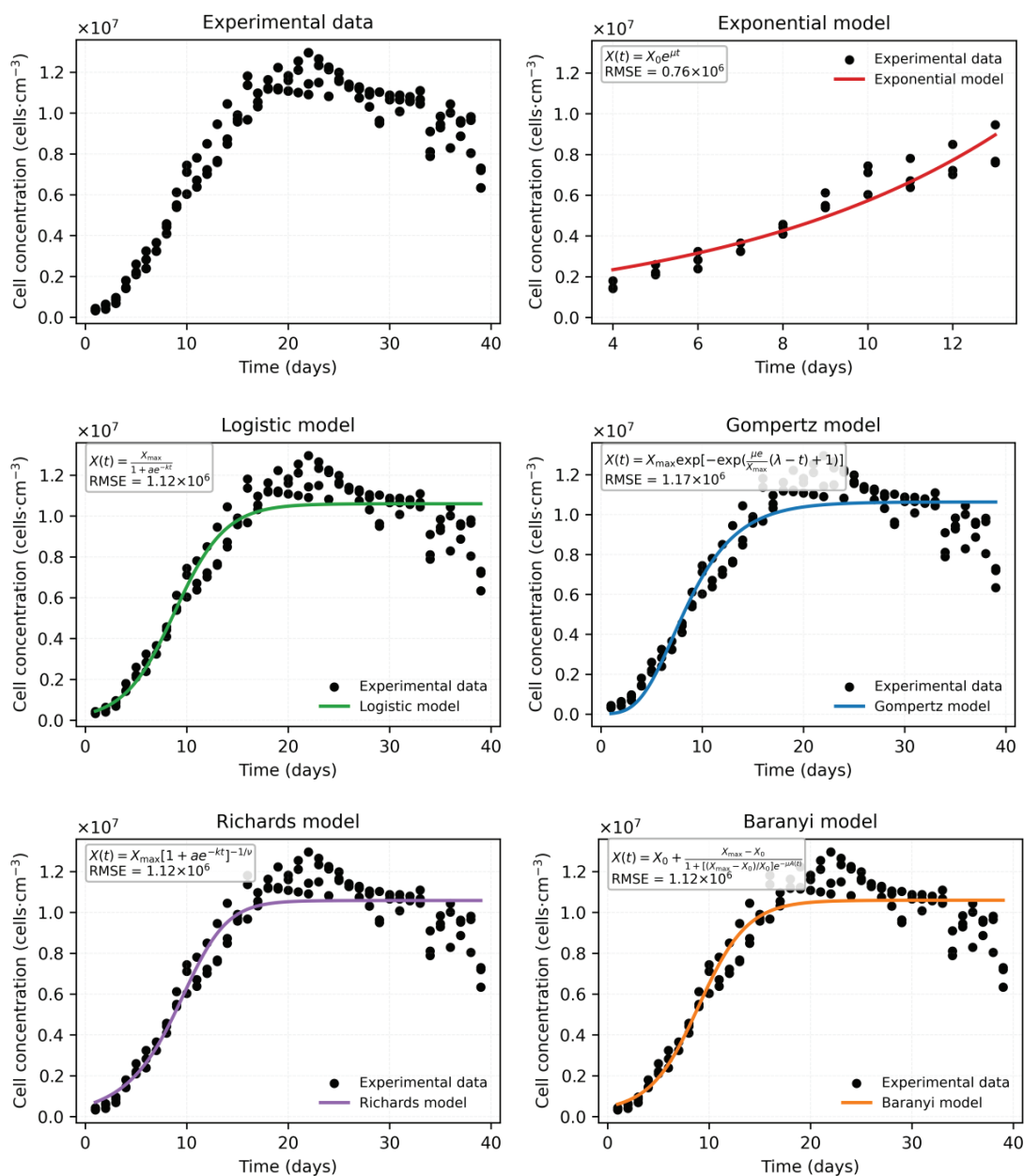


Fig. 2. Comparison of model-predicted cell density with experimental data; source: own study

(*RMSE*) as a descriptive goodness-of-fit metric. Values of *RMSE* were similar for the logistic ($1.123 \cdot 10^6$ cells·cm⁻³) and Richards ($1.116 \cdot 10^6$ cells·cm⁻³) models, confirming their comparable descriptive performance. The exponential model was fitted only to the exponential growth phase (days 4–13), as its assumptions are not valid during the deceleration and stationary phases. When applied to the entire dataset, the model exhibited a substantially higher *RMSE* ($6.094 \cdot 10^6$ cells·cm⁻³), confirming its limited suitability for describing the full growth trajectory (Tab. 4).

Table 4. Comparison of growth models for *Dunaliella* sp.

Model	Parameter (<i>k</i>)	<i>RMSE</i> (cells·cm ⁻³)	<i>BF</i>	<i>AICc</i>	$\Delta AICc$	<i>w_i</i>
Logistic	3	1.123e+06	1.010	3266.18	0.00	0.561
Richards	4	1.116e+06	1.042	3266.90	0.71	0.393
Baranyi	4	1.138e+06	1.000	3271.38	5.19	0.042
Gompertz	3	1.170e+06	0.870	3275.79	9.61	0.005
Exponential	2	6.094e+06	0.363	3657.77	391.59	<0.001

Explanations: *RMSE* = root mean square error, *BF* = bias factor, *AICc* = Akaike's information criterion corrected for small sample size, $\Delta AICc$ = difference in Akaike's information criterion corrected, *w_i* = Akaike weight.
Source: own study.

The experimental data revealed a classic sigmoidal growth pattern in *Dunaliella* sp., with a short lag phase and rapid exponential growth between days 2 and 20, followed by a plateau and a mild decline (Fig. 2). Akaike's information criterion corrected for small sample size (*AICc*), together with the difference in Akaike's information criterion corrected ($\Delta AICc$) and Akaike weights (*w_i*), was used as the primary basis for model comparison (Tab. 4). Comparison based on *AICc* indicated that the logistic model had the strongest statistical support (*AICc* = 3266.18; $\Delta AICc$ = 0.00; *w_i* = 0.561). The Richards model also showed substantial support ($\Delta AICc$ = 0.71; *w_i* = 0.393), indicating that both formulations are statistically plausible. The Gompertz model performed better than the Baranyi model, although its higher $\Delta AICc$ and lower Akaike weight indicated considerably weaker support than for the logistic and Richards models. In contrast, the exponential model exhibited the poorest performance, confirming its limited suitability for describing the complete growth trajectory.

To further assess systematic prediction bias, the bias factor (*BF*) was calculated for each model (Tab. 4). The Baranyi and logistic models exhibited *BF* values close to unity (1.000 and 1.010, respectively), indicating unbiased predictions on average. The Richards model showed a slight tendency towards overestimation (*BF* = 1.042), while the Gompertz and exponential models systematically underestimated the observed biomass concentrations (*BF* = 0.870 and 0.363, respectively). The combined analysis of graphical agreement (Fig. 2), goodness-of-fit metrics, information-theoretic criteria, and the bias factor consistently indicated that flexible sigmoidal models provided the most adequate description of *Dunaliella* sp. growth under batch cultivation. Among these, the logistic, Richards, and Baranyi models received the strongest statistical support, while the logistic model represented a simpler but still robust alternative.

The adequacy of the logistic model was further evaluated through residual analysis (Fig. 3). The residuals of the logistic

model were symmetrically distributed around zero and did not exhibit a systematic temporal structure, suggesting that the model assumptions of independence and structural adequacy were reasonably satisfied. No clear curvature or trend was observed across the time domain, indicating that the logistic formulation adequately captured the sigmoidal growth pattern, including the deceleration phase. The magnitude of the residuals remained relatively stable throughout the cultivation period, with no pronounced heteroscedasticity. This supports the assumption of approximately constant variance. The absence of clustering or

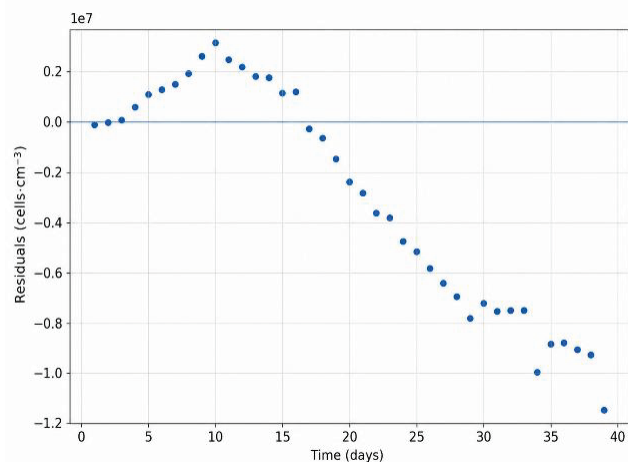


Fig. 3. Residual analysis of the logistic growth model; source: own study

systematic bias further corroborates the bias factor values close to unity and confirms that the model does not consistently overestimate or underestimate biomass concentration.

DISCUSSION

The comparative evaluation of five classical nonlinear growth models revealed substantial differences in their capacity to describe the complete batch growth trajectory of *Dunaliella* sp. As expected, the exponential model failed to represent growth deceleration and stationary-phase dynamics, confirming its limited applicability beyond the early exponential phase in closed cultivation systems (Baranyi and Roberts, 1994). This limitation stems from its assumption of a constant specific growth rate and the absence of environmental constraints. Such conditions are rarely maintained in batch microalgal cultures.

In contrast, sigmoidal models provided a more realistic description of the observed growth pattern. According to the information-theoretic analysis, the logistic model showed the strongest statistical support, achieving the lowest Akaike's information criterion corrected for small sample size (AIC_c) and the highest Akaike weight (w_i) (Burnham and Anderson, 2002). Although the Richards model produced a similar root mean square error ($RMSE$) and remained statistically plausible ($\Delta AIC_c < 2$), its additional shape parameter did not provide a sufficiently better fit to justify the increased complexity penalty. These findings underscore the importance of applying model-selection criteria that explicitly balance goodness-of-fit and parsimony rather than relying solely on descriptive statistics.

The biological interpretation of the deceleration phase further supports the adequacy of sigmoidal formulations. The reduction in specific growth rate observed after the exponential phase is consistent with progressive nutrient depletion, particularly nitrogen limitation, which restricts protein synthesis and cellular division. Increasing biomass concentration likely intensified self-shading effects, reducing effective light penetration and consequently limiting photosynthetic efficiency. Furthermore, halotolerant microalgae such as *Dunaliella* can undergo metabolic adjustments under stress conditions, reallocating resources towards protective metabolites rather than biomass accumulation. These physiological and environmental constraints collectively explain the transition towards the stationary phase and justify the use of growth models incorporating carrying capacity or asymmetry parameters.

The kinetic parameters estimated in this study are consistent with values reported in previous investigations of *Dunaliella* sp. and related microalgal systems (Halimi *et al.*, 2014; Sohrabi, Jazini and Shariati, 2019). Maximum specific growth rates ($\mu_{\max} \approx 0.25\text{--}0.30\text{ d}^{-1}$) fall within the ranges reported for batch cultivation under controlled laboratory conditions. Likewise, the estimated maximum cell densities ($X_{\max}$ on the order of $10^6\text{ cells}\cdot\text{cm}^{-3}$) are consistent with phototrophic growth in small-scale systems. A comparative summary of $\mu_{\max}$ and $X_{\max}$ values reported in selected studies is presented in Table 5.

advantageous when deceleration dynamics deviate from classical logistic behaviour (Wang, Liu and Huang, 2016). The Baranyi formulation, although biologically meaningful because of its explicit treatment of cellular adaptation (Sin *et al.*, 2024), did not outperform the logistic model under the present experimental conditions according to the AIC_c -based ranking.

From a bioprocess perspective, model selection has practical implications for scale-up and operational design. Reliable estimation of $\mu_{\max}$ and $X_{\max}$ supports the prediction of harvest timing, biomass productivity trends, and the optimisation of cultivation strategies. Parsimonious models such as the logistic formulation may provide greater robustness in industrial monitoring scenarios where parameter stability and interpretability are critical. Conversely, more flexible formulations may be required when cultivation conditions induce pronounced asymmetry or extended adaptation phases.

Despite the robustness of the modelling framework, certain limitations must be acknowledged. The study was conducted using a single laboratory strain identified at the genus level and evaluated under a single cultivation regime. Model fitting was performed on averaged biological triplicates rather than through replicate-level nonlinear modelling, which may limit the representation of biological variability. Additionally, the evaluated models do not explicitly incorporate decline or mortality kinetics, potentially affecting late-stage dynamics. Future studies should validate model rankings across multiple strains and environmental conditions and explore hierarchical or mixed-effects approaches to better capture biological variability.

Overall, the present findings demonstrate that information-theoretic model comparison provides a rigorous and reproducible basis for selecting growth models in microalgal kinetic studies. Under the evaluated conditions, the logistic model offers the most parsimonious and statistically supported representation of *Dunaliella* sp. growth dynamics. The Richards model remains a competitive alternative when greater flexibility is required.

Table 5. Comparative kinetic parameters ($\mu_{\max}$ and $X_{\max}$) reported for *Dunaliella* sp. and related studies

Study	Species	Cultivation conditions	Model	$\mu_{\max}$ (day^{-1})	$X_{\max}$ ($\text{cells}\cdot\text{cm}^{-3}$)
This study	<i>Dunaliella</i> sp.	batch, Conway medium, $25 \pm 1^\circ\text{C}$, $100\text{ }\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, 2% CO_2	logistic	0.27	$1.85 \cdot 10^6$
			Richards	0.30	$1.90 \cdot 10^6$
Halimi <i>et al.</i> (2014)	<i>Dunaliella tertiolecta</i> Butcher	batch, variable light intensity	Baranyi	0.32–1.07	no data
Sohrabi, Jazini and Shariati (2019)	<i>Dunaliella salina</i>	continuous/mixotrophic cultivation	no data	0.20–0.45	no data
Jesus and Filho (2010)	<i>Dunaliella salina</i> (Dunal) Teodoresco	mixotrophic room temperature with agitation	logistic	0.64 ± 0.09	$0.89 \cdot 10^6$

Explanations: $\mu_{\max}$, $X_{\max}$ as in Tab. 3.

Source: own study and literature data.

Mechanistically, the superior performance of the logistic model can be attributed to its capacity to represent resource-limited growth through a carrying-capacity term while maintaining minimal parameterisation. This structural simplicity enhances parameter identifiability and reduces the risk of overfitting. The Richards model, through its additional shape parameter, offers flexibility in capturing asymmetric growth curves, which may be

CONCLUSIONS

This study comparatively evaluated five classical nonlinear growth models for describing the batch cultivation of *Dunaliella* sp. under controlled laboratory conditions. Among the tested

formulations, the logistic model received the strongest statistical support based on Akaike's information criterion corrected for small sample size (AIC_c) and Akaike weights (w_i), indicating an optimal balance between descriptive accuracy and parameter parsimony. Although the Richards model exhibited a comparable root mean square error (RMSE) and remained statistically plausible ($\Delta AIC_c < 2$), its additional flexibility did not provide sufficient improvement to outweigh the complexity penalty imposed by the information-theoretic framework.

The Baranyi model adequately described the growth trajectory but showed lower statistical support, while the Gompertz and exponential models were less suitable for representing the complete sigmoidal profile observed in batch culture. Residual diagnostics confirmed the absence of systematic bias in the logistic formulation, reinforcing its structural adequacy.

Beyond descriptive performance, these findings have direct implications for microalgal bioprocess modelling and scale-up strategies. The identification of a statistically supported and parsimonious kinetic formulation enhances predictive reliability for harvest timing, productivity estimation, and parameter stability in simulation frameworks. In large-scale cultivation systems, where operational robustness and model interpretability are critical, the logistic model provides a stable basis for integration into process optimisation, monitoring, and control strategies. Furthermore, the application of information-theoretic criteria establishes a reproducible framework for evaluating growth models under varying environmental and operational conditions. This supports future efforts to predict *Dunaliella* sp. growth dynamics beyond laboratory-scale experiments.

AUTHOR CONTRIBUTIONS

1st Author: study design, statistical analysis, data interpretation, manuscript preparation; 2nd Author: study design, manuscript preparation, literature search; 3rd Author: data interpretation; 4th Author: statistical analysis; 5th Author: data collection.

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CONFLICT OF INTERESTS

All authors declare that they have no conflict of interests.

REFERENCES

- Abbas, M., Ni, L. and Du, C. (2024) "Assessment of hydrocarbon degradation capacity and kinetic modeling of *Chlorella vulgaris* and *Scenedesmus quadricauda* for crude oil phycoremediation under mixotrophic conditions," *International Journal of Phytoremediation*, 26(12), pp. 1914–1922. Available at: <https://doi.org/10.1080/15226514.2024.2361318>.
- Abbas, M., Ni, L. and Du, C. (2025) "Kinetic modelling and salinity tolerance in *Chlorella vulgaris* and *Desmodesmus communis* (Chlorophyta): Insights into differential growth responses," *Aquatic Ecology*, 59(1), pp. 145–158. Available at: <https://doi.org/10.1007/s10452-024-10153-y>.
- Baldiris-Navarro, I. et al. (2023) "Application of *Nannochloris* sp. for landfill leachate biotreatment and lipids production," *Indonesian Journal of Chemistry*, 23(5), pp. 1456–1462. Available at: <https://doi.org/10.22146/ijc.82483>.
- Baranyi, J., and Roberts, T.A. (1994) "A dynamic approach to predicting bacterial growth in food," *International Journal of Food Microbiology*, 23(3–4), pp. 277–294. Available at: [https://doi.org/10.1016/0168-1605\(94\)90157-0](https://doi.org/10.1016/0168-1605(94)90157-0).
- Bhat, O., Unpaprom, Y., and Ramaraj, R. (2024) "Exploring the impact of light-emitting diodes and pH on microalgae *Spirulina* biomass production optimization using a central composite design," *Biomass Conversion and Biorefinery*, 14(8), pp. 9839–9847. Available at: <https://doi.org/10.1007/s13399-023-04666-z>.
- Burnham, K.P. and Anderson, D. R. (2002) *Model selection and multimodel inference: A practical information-theoretic approach*. 2nd edn. New York: Springer.
- Esteves, A.F. et al. (2024). "Comparative assessment of microalgal growth kinetic models based on light intensity and biomass concentration," *Bioresource Technology*, 394, 130167. Available at: <https://doi.org/10.1016/j.biortech.2023.130167>.
- Gavin, H.P. (2026) *The Levenberg–Marquardt algorithm for nonlinear least squares curve-fitting problems*. Durham, NC: Duke University. Available at: <https://people.duke.edu/~hpgavin/ce281/lm.pdf> (Accessed: July 1, 2025).
- Halmi, M.I.E. et al. (2014) "Evaluation of several mathematical models for fitting the growth of the algae *Dunaliella tertiolecta*," *Asian Journal of Plant Biology*, 2(1), pp. 1–6. Available at: <https://doi.org/10.54987/ajpb.v2i1.81>.
- Hecke van, T. (2017) "The Levenberg-Marquardt method to fit parameters in the Monod kinetic model," *Journal of Statistics and Management Systems*, 20(5), pp. 953–963. Available at: <https://doi.org/10.1080/09720510.2017.1325090>.
- Jesus, S.S. and Filho, R.M. (2010) "Modeling growth of microalgae *Dunaliella Salina* under different nutritional conditions," *American Journal of Biochemistry and Biotechnology*, 6(4), pp. 279–283. Available at: <https://doi.org/10.3844/ajbbsp.2010.279.283>.
- Kassim, M.A., Ramli, S.H. and Meng, T.K. (2022) "Analysis of microalgal growth kinetic model and carbohydrate biosynthesis cultivated using agro-industrial waste residuals as carbon source," *Preparative Biochemistry & Biotechnology*, 52(5), pp. 514–524. Available at: <https://doi.org/10.1080/10826068.2021.1969576>.
- Laizu, S.A., Hassan, L.H.S. and Ahmad, I. (2025) "Chapter 7: Open and closed photobioreactors for microalgal cultivation and wastewater treatment," in I. Ahmad and N. Abdullah (eds.) *Nature-based technologies for wastewater treatment and bioenergy production*, 121. Available at: https://doi.org/10.2166/9781789064100_0121.
- Ling, K.S., Jamaian, S.S. and Zulkifli, F.H. (1988) "Estimation of the microalgae growth using the optimization approach," *Journal of*

- Physics: Conference Series*, 1988(1), 012017. Available at: <https://doi.org/10.1088/1742-6596/1988/1/012017>.
- Ma, X. and Jian, W. (2023) "Growth conditions and growth kinetics of *Chlorella vulgaris* cultured in domestic sewage," *Sustainability*, 15(3), 2162. Available at: <https://doi.org/10.3390/su15032162>.
- Mehariya, S. *et al.* (2021) "Integrated approach for wastewater treatment and biofuel production in microalgae biorefineries," *Energies*, 14(8), 2282. Available at: <https://doi.org/10.3390/en14082282>.
- Mohd, N., Yasin, N.H.M. and Takriff, M.S. (2021) "Predictive growth model of indigenous green microalgae (*Scenedesmus* sp. UKM9) in palm oil mill effluent (POME)." *IOP Conference Series: Materials Science and Engineering*, 1051(1), 5th International Conference on Advanced Technology and Applied Sciences 2020 ICATAS 2020 in conjunction with the 6th Malaysia-Japan Joint International Conference 2020 MJJIC 2020 (ICATAS-MJJIC 2020), 012070. Available at: <https://doi.org/10.1088/1757-899X/1051/1/012070>.
- Pérez-Guzmán, S.M. *et al.* (2024) "Kinetics of obtaining microalgal biomass and removal of organic contaminants in photobioreactors operated with microalgae – Study case: Treatment of wastewater from a poultry slaughterhouse," *Water*, 16(11), 1558. Available at: <https://doi.org/10.3390/w16111558>.
- Piyatilleke, S. *et al.* (2024) "Microalgal biofuels: Challenges and prospective in the framework of circular bioeconomy," *Energy Nexus*, 17, 100338. Available at: <https://doi.org/10.1016/j.nexus.2024.100338>.
- Ramírez Duque, J.L.R. (2017) "Viabilidad en la producción de biomasa microalgal a partir de fotobioreactores solares en el Valle del Cauca, Colombia [Feasibility of microalgal biomass production from solar photobioreactors in the Cauca Valley, Colombia]," *Revista de Investigación Agraria y Ambiental*, 8(2), pp. 127–140. Available at: <https://doi.org/10.22490/21456453.2039>.
- Sin, P.Y. *et al.* (2024) "Probiotic growth pattern and physicochemical evaluation of water kefir fermentation, *Malaysian Applied Biology*, 53(2), pp. 21–30. Available at: <https://doi.org/10.55230/mabjournal.v53i2.2742>.
- Sohrabi, D., Jazini, M.H. and Shariati, M. (2019) "Mixotrophic cultivation of *Dunaliella salina* on crude glycerol obtained from calcinated fatty acid production process," *Russian Journal of Marine Biology*, 45, pp. 470–480. Available at: <https://doi.org/10.1134/S1063074019060105>.
- Sousa, C.A. *et al.* (2021) "Microalgae-based bioremediation of wastewaters – Influencing parameters and mathematical growth modelling," *Chemical Engineering Journal*, 425, 131412. Available at: <https://doi.org/10.1016/j.cej.2021.131412>.
- Tambat, V.S. *et al.* (2025) "Advancing sustainable astaxanthin-lipid biorefineries: Robust two-stage phytohormone-driven bioprocess in *Chromochloris zofingiensis*," *Bioresource Technology Reports*, 29, 102022. Available at: <https://doi.org/10.1016/j.biteb.2025.102022>.
- Tan, X. *et al.* (2025) "Adaptation responses of *Dunaliella* sp. and its natural symbiotic bacteria to sulfamethoxazole," *Environmental Technology & Innovation*, 38, 104171. Available at: <https://doi.org/10.1016/j.eti.2025.104171>.
- Wang, J. and Guo, X. (2024) "The Gompertz model and its applications in microbial growth and bioproduction kinetics: Past, present and future," *Biotechnology Advances*, 72, 108335. Available at: <https://doi.org/10.1016/j.biotechadv.2024.108335>.
- Wang, X., Liu, S. and Huang, Y. (2016) "A study on the rapid parameter estimation and the grey prediction in Richards model," *Journal of Systems Science and Information*, 4(3), pp. 223–234. Available at: <https://doi.org/10.21078/JSSI-2016-223-12>.
- Yan, Z. *et al.* (2021) "Adaptive Levenberg–Marquardt algorithm: A new optimization strategy for Levenberg–Marquardt neural networks," *Mathematics*, 9, 2176. Available at: <https://doi.org/10.3390/math9172176>.